Modular Arithmetics

<u>Definition</u>. For $n \in \mathbb{N}$, $n \geq 1$, the set $\mathbb{Z}_n$ or $\mathbb{Z}/n\mathbb{Z}$ is the set of remainders when dividing by n.

$$\mathbb{Z}_n = \{0, \ldots, n-1\}.$$

Arithmetic on $\mathbb{Z}_n$: $+, -, \times$ allowed.

If $a, b \in \mathbb{Z}_n$, $b \neq 0$, we can define a/b as the number $x \in \mathbb{Z}_n$ such that $bx \equiv a \pmod{n}$ if it exists.

Fields

<u>Definition (Fields)</u>. A field $(F, +, \cdot)$ is a set F with an "addition" $+$ and a "multiplication" $\cdot$ with the following properties: $\forall a, b, c \in F$,

1. Associativity: $(a+b)+c = a+(b+c)$
   $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. Commutativity: $a+b = b+a$, $a \cdot b = b \cdot a$.
3. Distribution: $a \cdot (b+c) = a \cdot b + a \cdot c$

4. Identities: There is $0 \in F$ and $1 \in F$, and such that $a + 0 = a$, $a \cdot 1 = a$.

5. Inverses: There is $-a \in F$ for all $a \in F$ with $a + (-a) = 0$. There is $a^{-1} \in F$ for all $a \in F$, $a \neq 0$, with $a \cdot a^{-1} = 1$.

In a field, we can always subtract and divide.

$$a - b := a + (-b),$$
$$a/b := a \cdot b^{-1}.$$

<u>Theorem</u>. $\mathbb{Z}_n$ is a field if and only if n is a prime.

Intro to Groups

<u>Definition.</u> A group $(G, *)$ is a set G and an operation $*$ with the following properties for all $a, b, c \in G$.

1. Associativity: $(a * b) * c = a * (b * c)$
2. Identity: There is an element $e \in G$ such that $a * e = e * a = a$, $\forall a \in G$.

3. Inverses: For all $a \in G$ there is an element $b \in G$ such that $a * b = b * a = e$.

Note: We do not assume commutativity or abelian, i.e. $\forall a, b \in G$, $a * b = b * a$.